Anuj Kumar Yadav

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Education

Swiss Federal Institute of Technology in Lausanne (EPFL) — **Lausanne, Switzerland**
PhD in Computer and Communication Sciences — *2022–2027 (Expected)*
Advisors: Prof. Yanina Shkel and Prof. Michael Gastpar

Stanford University — **California, USA**
Visiting Student, Electrical Engineering — *Sept. 2025–Feb. 2026*
Mentor: Prof. Ayfer Özgür

University of Bristol — **England, UK**
Visiting Student, Mathematics — *2021*
Mentor: Prof. Sidharth (Sid) Jaggi

Indian Institute of Technology (IIT) Patna — **Patna, India**
B.Tech. in Electrical Engineering — *2018–2022*
Advisors: Prof. Amitalok Budkuley and Prof. Preetam Kumar

Research Interests

Information-theoretic and **probabilistic methods** for **statistics**, **learning**, and **privacy**.
Current focus on **probability couplings** and **private statistical inference**.

Industry Experience

Nokia Bell Labs — Research Scientist Intern — **Stuttgart, Germany**
Continuous health monitoring in passive optical networks (PONs) — *June 2024–Aug. 2024*
Mentors: Dr. Christoph Füllner and Dr. René Bonk

- I worked on in vivo health monitoring for 50-gigabit-capable passive optical networks (PONs) using digital signal processing, statistical, and machine learning techniques. In particular, I designed non-parametric change-point detection algorithms to track parameter changes and detect faults in the PON system, and devised ML-based classification algorithms to localize those faults.

Publications

Remark: Publications marked **(A2Z)** use alphabetical author order; * indicates equal contributions.

Preprints

1. A. K. Yadav and Y. Y. Shkel, "Minimum Rényi Entropy Couplings and Functional Representations," 2026. [*In preparation*]
2. A. K. Yadav, C. Cadir, Y. Shkel, and M. Gastpar, "Locally Private Parametric Methods for Change-Point Detection," 2026. [*Paper*] [*Under review*]
3. A. K. Yadav and Y. Y. Shkel, "Couplings and Entropic Inequalities on the Majorization Lattice," *IEEE Transactions on Information Theory*, 2026. [*Paper*] [*Under review*]
4. S. Chandak, A. K. Yadav, A. Özgür, and N. Bambos, "Heavy-Tailed and Long-Range Dependent Noise in Stochastic Approximation: A Finite-Time Analysis," *IEEE Transactions on Automatic Control*, 2026. [*Paper*] [*Under review*]

Conferences and Workshops

1. A. K. Yadav and Y. Shkel, "Geometry of Rényi Entropy on the Majorization Lattice," *2026 IEEE Information Theory Workshop (ITW) Conference*, Tempe (To appear). [*Paper*]
2. A. K. Yadav, C. Cadir, Y. Shkel, and M. Gastpar, "Locally Private Parametric Methods for Change-Point Detection," *2026 NeurIPS TS-LIMITS Workshop*, Paris (Non-archival). [*Paper*]

3. A. K. Yadav, D. Song, Y. Shkel, and A. Özgür, "Log-Likelihood Loss for Semantic Compression," *2026 IEEE International Symposium on Information Theory (ISIT)*, Guangzhou. [*Paper*]
4. A. K. Yadav and Y. Y. Shkel, "Approximation Guarantees for Minimum Rényi Entropy Functional Representations," *2025 IEEE International Symposium on Information Theory (ISIT)*, Ann Arbor. [*Paper*]
5. A. K. Yadav, M. Mamindlapally, and A. J. Budkuley, "Leveraging Spherical Codes for Commitment over Gaussian UNCs," *2025 International Workshop on Code-Based Cryptography (CBCrypto), Eurocrypt*, Madrid (Non-archival). [*Paper*]
6. C. Füllner, A. K. Yadav, and R. Bonk, "DSP-Assisted In-Vivo Health Monitoring for ≥ 50G PON," *2025 Optical Fiber Communication Conference (OFC)*, San Francisco. [*Paper*]
7. A. K. Yadav, M. Mamindlapally, and A. J. Budkuley, "Wiretapped Commitment over Binary Channels," *2024 IEEE International Symposium on Information Theory (ISIT)*, Athens. [*Paper*]
8. **(A2Z)** Y. Y. Shkel* and A. K. Yadav*, "Information Spectrum Converse for Minimum Entropy Couplings and Functional Representations," *2023 IEEE International Symposium on Information Theory (ISIT)*, Taipei. [*Paper*]
9. **(A2Z)** A. J. Budkuley*, P. Joshi*, M. Mamindlapally*, and A. K. Yadav*, "On the (Im)possibility of Commitment over Gaussian Unfair Noisy Channels," *2023 IEEE International Symposium on Information Theory (ISIT)*, Taipei. [*Paper*]
10. **(A2Z)** A. J. Budkuley*, P. Joshi*, M. Mamindlapally*, and A. K. Yadav*, "Commitment over Unreliable Noisy Channels: When Awareness Meets Control," *2022 IEEE Information Theory Workshop (ITW) Conference*, Mumbai. [*Paper*]
11. A. K. Yadav, M. Alimohammadi, Y. Zhang, A. J. Budkuley, and S. Jaggi, "New Results on AVCs with Omniscient and Myopic Adversaries," *2022 IEEE International Symposium on Information Theory (ISIT)*, Espoo. [*Paper*]
12. A. K. Yadav, M. Mamindlapally, P. Joshi, and A. Budkuley, "On Commitment over General Compound Channels," *2022 IEEE International Conference on Communication Systems & Networks (COMSNETS)*, Bangalore. [*Paper*]
13. **(A2Z)** A. J. Budkuley*, P. Joshi*, M. Mamindlapally*, and A. K. Yadav*, "On the Commitment Capacity of Reverse Elastic Channels," *2021 IEEE Information Theory Workshop (ITW) Conference*, Kanazawa. [*Paper*]
14. A. K. Yadav, M. Mamindlapally, A. J. Budkuley, and M. Mishra, "Commitment over Compound Binary Symmetric Channels," *2021 IEEE National Conference on Communications (NCC)*, Kanpur. [*Paper*]
15. M. Mamindlapally, A. K. Yadav, M. Mishra, and A. J. Budkuley, "Commitment Capacity under Cost Constraints," *2021 IEEE International Symposium on Information Theory (ISIT)*, Melbourne. [*Paper*]

Journals

1. A. K. Yadav and P. Kumar, "Oblivious Transfer over Compound Binary Erasure Channels," *IEEE Communications Letters*, 2022. [*Paper*]
2. **(A2Z)** A. J. Budkuley*, P. Joshi*, M. Mamindlapally*, and A. K. Yadav*, "On Reverse Elastic Channels and the Asymmetry of Commitment Capacity under Channel Elasticity," *IEEE Journal on Selected Areas in Communications*, 2022. [*Paper*] [*Blog by the Editor-in-Chief of IEEE JSAC*]

Posters & Talks

1. Invited talk on "Minimum Rényi Entropy Couplings and Their Applications" at *Inria*, Sophia Antipolis, Aug. 2026. [*Slides*] [*Video*]
2. Poster presentation on "Log-Likelihood Loss for Semantic Compression" at the *2026 Information Theory in Modern Science Workshop*, Okinawa. [*Poster*]
3. Talk on "Minimum Rényi Entropy Couplings (and Applications)" at the *2025 INFORMS Applied Probability Conference (APS)*, Atlanta. [*Slides*]
4. Poster presentation on "Minimum Rényi Entropy Couplings" at the *2025 IEEE European School of Information Theory (ESIT)*, Ancona, Italy. [*Poster*]

5. Presentation on "In-vivo Health Monitoring for Passive Optical Networks" at *Nokia Bell Labs*, Murray Hill, NJ, USA, Aug. 2024. [*Slides*]
6. Invited talk on "Commitment Capacity under Cost Constraints" at *Laboratoire ETIS*, CNRS, Jan. 2022. [*Slides*]
7. Poster presentation on "Role of Costs in Commitment over Noisy Channels" at the *2021 IEEE North American School of Information Theory (NASIT)*, UBC, Canada. [*Poster*]
8. Poster presentation on "Rate Positivity for Arbitrarily Varying Channels" at the *2021 Croucher Summer Course in Information Theory (CSCIT)*, Hong Kong. [*Abstract*] [*Poster*]

Academic and Research Activities

Teaching Assistant

- **COM-417**: Advanced Probability and Applications (Spring 2024, Fall 2024).
- **COM-302**: Principles of Digital Communications (Spring 2025).
- **COM-202**: Signal Processing (Fall 2023, Spring 2026).
- **EE-205**: Signals and Systems (Spring 2023).

Mentorship

- Master's project mentor: *Correlated sampling under bounded communication complexity* (Fall 2026).
- Master's project mentor: *Infinite divisibility of information* (Spring 2024).
- RAMP mentor and EDIC buddy for EPFL PhD applicants and students.

Peer Review

- IEEE Transactions on Information Theory (2024–present).
- IEEE Transactions on Communications (2025–present).
- AAAI, AAAI Undergraduate Consortium (2027).
- AISTATS (2027).
- NeurIPS (2026).
- ICML (2026).
- IEEE ISIT (2023–2026).
- IEEE ITW (2022–2026).

Awards & Achievements

ICML Gold Reviewer Award *May 2026*

- Recognized among the top 25% of reviewers for high-quality peer reviews at the International Conference on Machine Learning (ICML) 2026.

Swiss National Science Foundation (SNSF) Mobility Grant *April 2025*

- Received the Swiss National Science Foundation Mobility grant for PhD students to conduct research in information theory, applied probability, and statistics at Stanford University, USA.

Institute Proficiency Award, IIT Patna *Dec. 2022*

- Received the Institute Proficiency Award for the best bachelor's thesis in my cohort in the 2021–2022 academic year, titled "Secure Two-Party Computation: Information-Theoretic Oblivious Transfer."

IEEE ITSoc D&I Undergraduate Scholarship *June 2021*

- Awarded IEEE Information Theory Society (ITSoc) membership and the opportunity to attend ISIT 2021 and various ITSoc summer schools virtually, based on my undergraduate research in information theory.

Prime Minister's Scholarship Scheme (PMSS) *2018–2022*

- Received a PMSS scholarship from the Ministry of Home Affairs, Government of India, for four consecutive years based on academic merit during my bachelor's studies at IIT Patna.